

Advanced (Habanero)

- Q1.** What is the unit rate of $\frac{24 \text{ km}}{4 \text{ hours}}$?
(A) 6 km/h
(B) 8 km/h
(C) 4 km/h
(D) 12 km/h
(E) 3 km/h
- Q2.** If y varies directly with x , and $y = 12$ when $x = 4$, what is y when $x = 6$?
(A) 12
(B) 16
(C) 18
(D) 20
(E) 24
- Q3.** A car travels 250 miles in 5 hours. What is its average speed in miles per hour?
(A) 45 mph
(B) 50 mph
(C) 55 mph
(D) 60 mph
(E) 50 mph
- Q4.** What is the unit rate of $\frac{36 \text{ inches}}{3 \text{ feet}}$?
(A) 4 in/ft
(B) 12 in/ft
(C) 9 in/ft
(D) 1 in/ft
(E) 6 in/ft
- Q5.** A worker can complete a task in 8 hours. If two workers work together, how many hours will it take to complete the task?
(A) 2 hours
(B) 4 hours
(C) 6 hours
(D) 5 hours
(E) 3 hours
- Q6.** If y varies inversely with x , and $y = 9$ when $x = 3$, what is y when $x = 9$?
(A) 1
(B) 3
(C) 5
(D) 7
(E) 9
- Q7.** A bicycle travels 18 miles in 2 hours. How many miles will it travel in 5 hours?
(A) 30 miles
(B) 40 miles
(C) 45 miles
(D) 50 miles
(E) 55 miles
- Q8.** If it takes 5 machines 5 hours to complete a task, how many hours will it take 10 machines to complete the same task?
(A) 1 hour
(B) 2 hours
(C) 2.5 hours
(D) 5 hours
(E) 10 hours
- Q9.** A factory produces 480 units in 8 hours. If production is increased by 25%, how many units will be produced in 10 hours? Show your work and explain your reasoning.
- Q10.** A water tank can be filled by two pipes. Pipe A can fill the tank in 6 hours, and pipe B can fill it in 8 hours. How many hours will it take to fill the tank if both pipes are used together? Explain your solution and provide calculations.

Open Ended Questions (FRQ)

- Q9.** A factory produces 480 units in 8 hours. If production is increased by 25%, how many units will be produced in 10 hours? Show your work and explain your reasoning.
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Chef's Tips

- Tip 1: Ensure students understand the concept of unit rates before proceeding to more complex problems.
- Tip 2: Remind students to read each question carefully and choose the correct units for their answers.
- Tip 3: Encourage students to use diagrams or graphs to help model direct and inverse variation problems when necessary.